MINING DEALER BEST PRACTICE

# Sensor Fault Diagnosis Simulator Device

**Maintenance and Repair** | Application | Component Life Management | Component Renewal (CRC) | MARC Management






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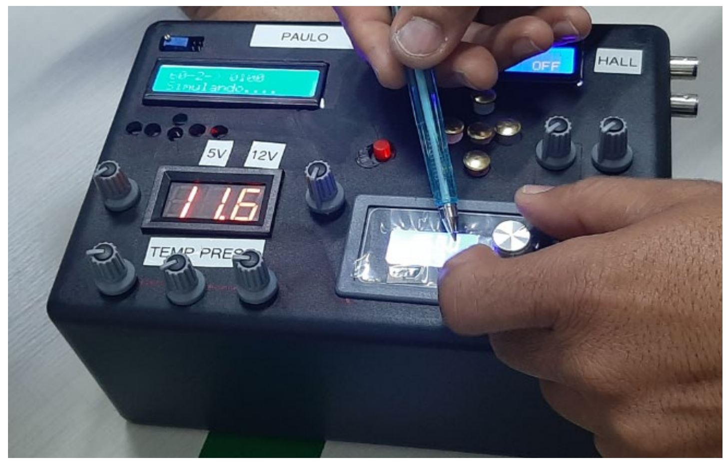
## 1.0 Introduction

This Best Practice developed by Sotreq provides an auxiliary diagnostic tool option to correct the fault or simulate pressures in less time, around 2 minutes, making the equipment less maintenance time, improving physical availability and consequently increasing production, the model is configurable to work on different Caterpillar models as needed.

The diagnostic simulator device checks if the fault occurs in:

• Sensor
• Harness
• ECM (Electronic Control Module)

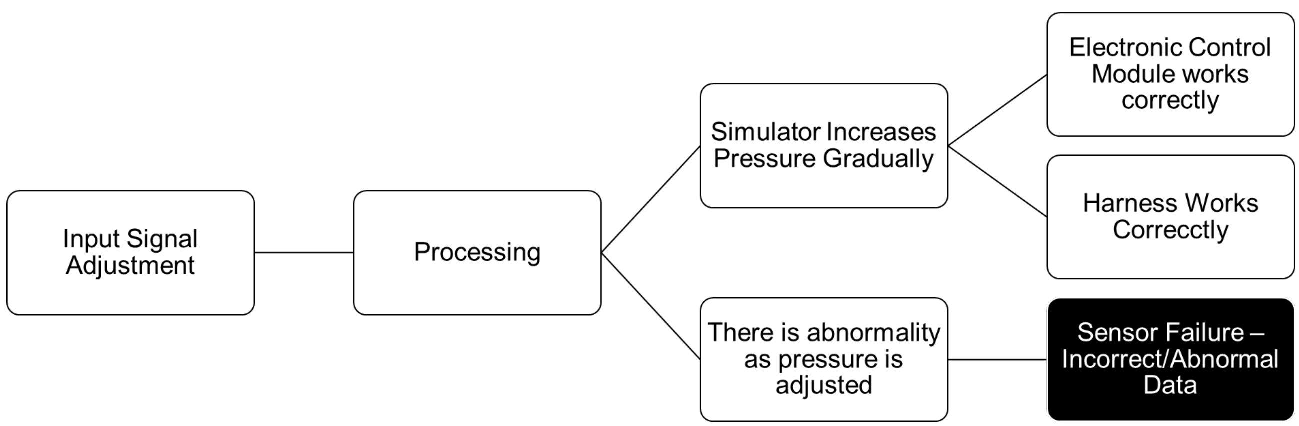




## 2.0 Best Practice Description

It is common in workshops that there is some delay in obtaining an electrical diagnosis due to factors such as lack of resources, skill, and lack of knowledge of the maintenance technician. In most cases, the diagnosis is not concluded with assertiveness and/or the customer also allows the equipment to return to operation with a failure so as not to impact the production of ore loading.

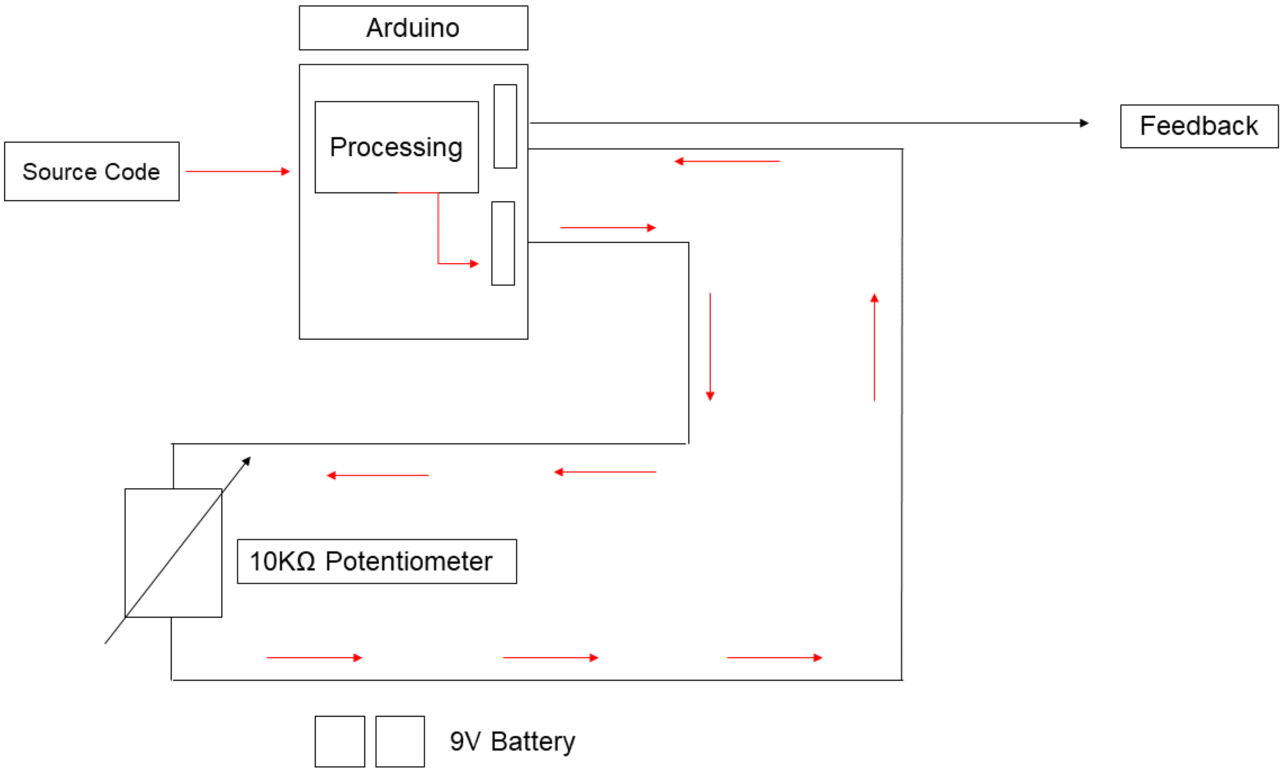
When the problem is not completely resolved, the client is directly impacted on its indicators such as DF, MTBF and MTTR. As an example: A diesel engine oil pressure sensor diagnostic can range from 01 to 04 hours depending on who is responsible for execution and based on what resources are available. If the shutdown occurs with the equipment undergoing corrective maintenance, the event will result in an increase in the meantime to repair (MTTR), and if the problem is not effectively solved, there will be a decrease in the mean time between failures (MTBF) as the equipment will certainly stop in corrective maintenance again in a short period of time for the same reason. There will also be a decrease in the indicator related to physical availability (DF) because the equipment will be at a standstill in maintenance.

The project used the Arduino® platform as the basis for its control system. Arduino® is an Open-Source platform for electronic prototyping that can be used with its own programming language developed for use with your hardware, also using its own editor of codes: The Arduino Software, which is an integrated development environment, with relatively simple command lines and few steps, the insertion of parameters related to pressure simulation (increase and decrease) can be done by connecting the equipment to the sensor allegedly damaged, then the input of information into the system occurs through potentiometers that allow the adjustment of the engine oil pressure parameters, the device's motherboard receives, processes and displays the data on the screen of a computer previously connected to the device, the software screen displays: Engine oil pressure in psi (With its improper operating status, if detected) engine oil level status, atmospheric pressure and accelerometer position, the advantage of using an Open-Source platform, specifically in the case of this project, is justified taking into account the very low cost compared to similar proprietary technologies, which makes it possible to replicate and implement the project more easily in the various maintenance and repair processes routinely carried out by the teams, in addition to allowing easy replacement of parts/components relatively quickly.

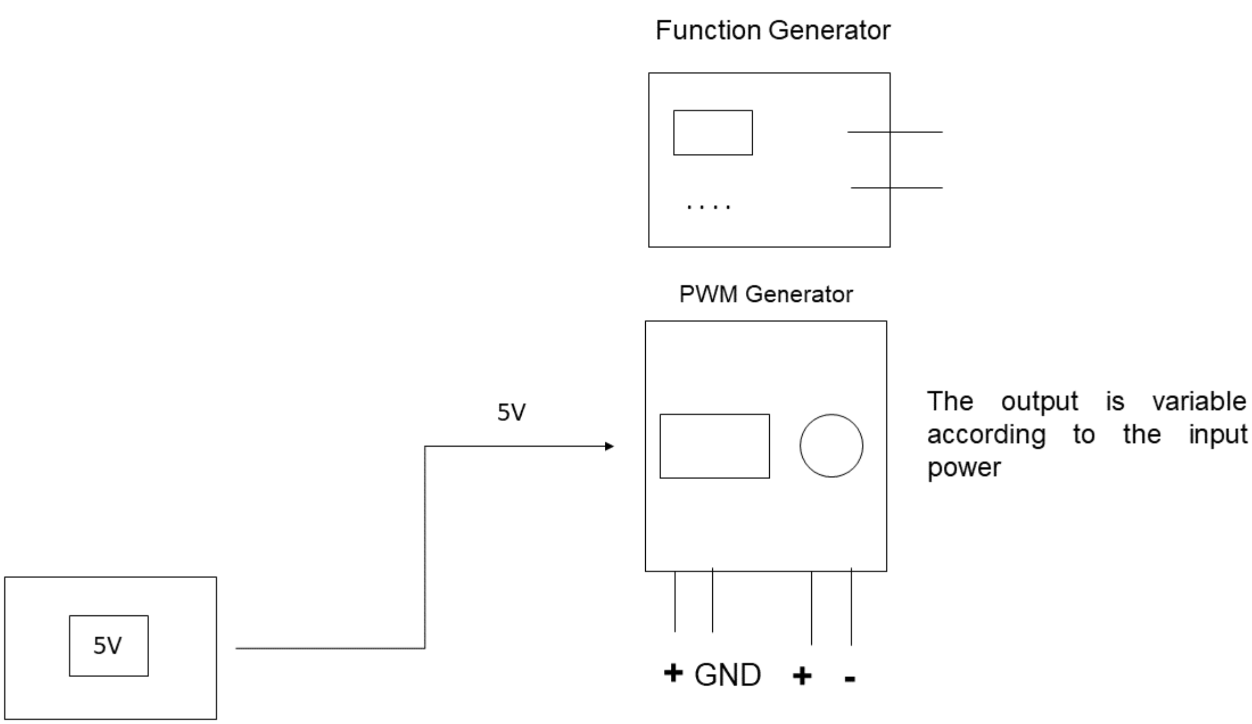



Process map exemplifying the failure detection performed by the system, in case of abnormality, the detection of incorrect functioning of the sensor, also allows to indirectly verify the proper functioning of other associated devices present in the equipment.

The functioning of the electronic system is outlined below with the representation of its constituents and operating flow (indicated by the red arrows) from the input of information, processing and feedback, which consists of the information displayed on the screen, the operating process is also exemplified. of the functions generator and the PWM generator, in addition to the identification of the components of each system.




Simplified layout for device operation, showing components.

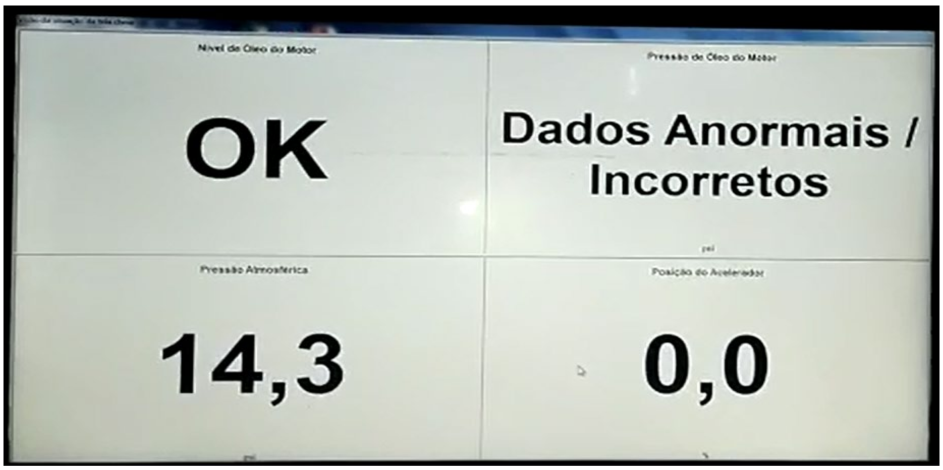



Simplified layout for Function Generator and PWM Generator.

## 3.0 Implementation Steps

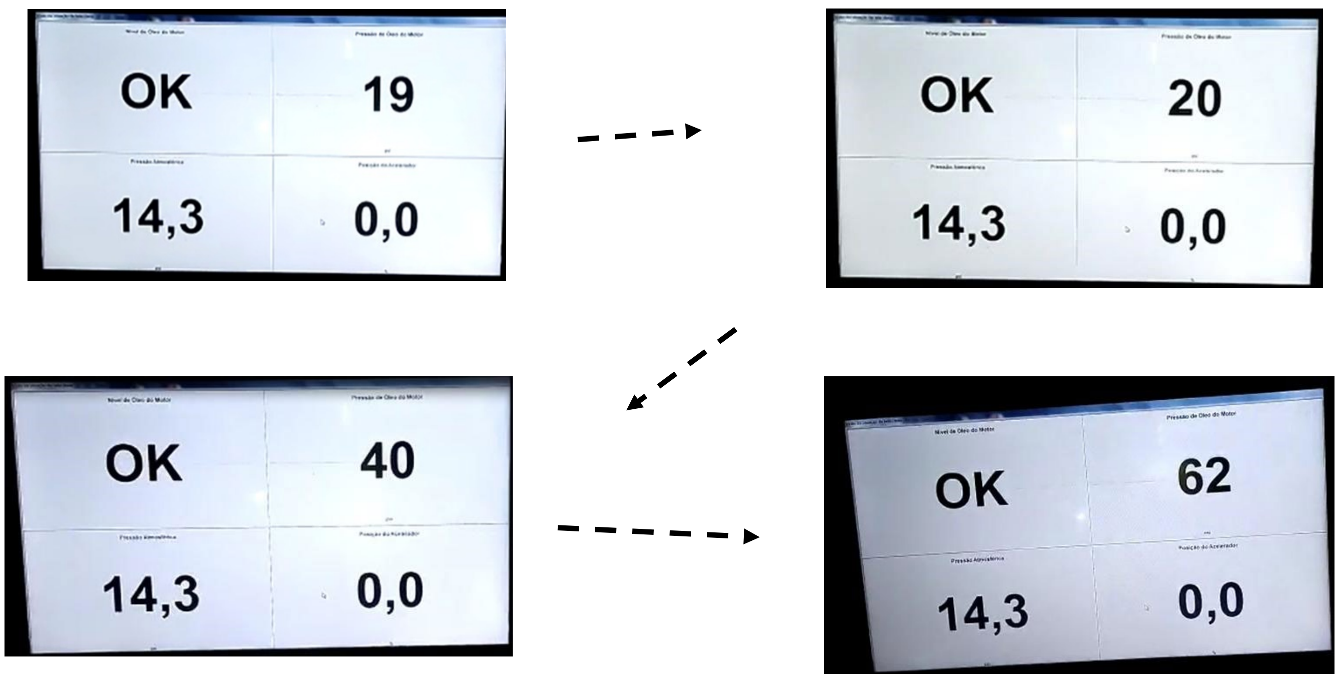
This process offers tangible benefits to customers and also to the Group's resellers, as its ultimate objective is to improve fleet productivity by reducing machine downtime in maintenance. This will improve your KPIs while bringing more reliability to the equipment and the process as a whole.

3.1 - Disconnect the sensor with the possible failure of the harness to which the first quadrant is connected informs the failure in question.




3.2 Disconnect the sensor with the possible failure of the harness to which the first quadrant is connected informs the failure in question.

3.3 When turning on the simulator, on the potentiometer, simulate the pressure of the sensor. If the pressure increases as it is manually graduated, this indicates that the harness and ECM are normal, indicating a failure in the sensor.


Simulation of pressure increase indicating correct operation of the harness and ECM (Electronic Control Module) This simulation makes possible the immediate detection of failure in the sensor.


Device working in the configuration step, with cables required for connections.

## 4.0 Benefits

- Decrease in the time needed to assertively carry out a failure diagnosis
- Reduction in Maintenance Time
- Simple construction tool, with the potential to circumvent any budget restrictions for its development.
- Easy usability makes the adoption of the device possible by technicians with less experience or procedural knowledge about technical fault investigation.

## 5.0 Resources Required

Table with resources needed to build the device.

| Item | Qty | Type | Description |
|---|---|---|---|
| 1 | 1 | Unit | Arduino |
| 2 | 1 | Unit | Function Generator |
| 3 | 1 | Unit | PWM Generator |
| 4 | 5 | Unit | Potentiometer |
| 5 | 2 | Unit | 9V Battery |
| 6 | 1 | Unit | LM 2904N Integrated Circuit |
| 7 | 1 | Unit | Printed Circuit Broad |
| 8 | 1,5 | m | 1,5 mm Wire |
| 9 | 1 | Unit | Project Box |
| 10 | 5 | Unit | Knob |
| 11 | 4 | Unit | Resistor |
| 12 | 6 | Unit | Audio and Video Connector |
| 13 | 2 | m | Microphone Cable |
| 14 | 1 | Unit | Alligator Clamp |

## 6.0 Supporting Attachments / References



Project box measurements for component allocation: Arduino, Function Generator, PWM Generator, Voltmeter and Potentiometers.

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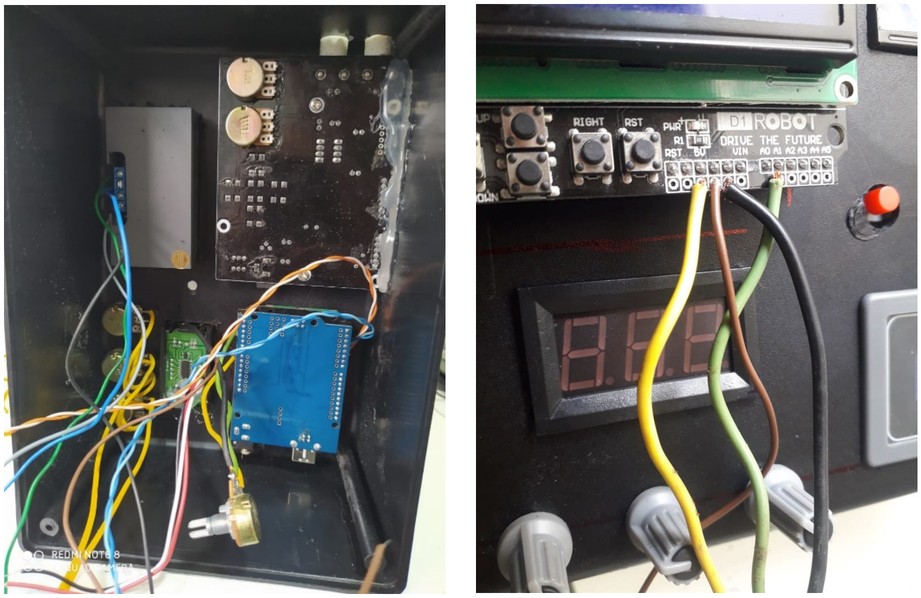

Connecting the Potentiometers to the Arduino, Function Generator and PWM Generator

More information:

- Perform system energy supply with 02 (two) 9V batteries.
- Connect the signal outputs to the DB9 connector.
- Carry out the manufacture of cables for connection between the Simulator Box and the harness.

## 7.0 Related Best Practices

The project is entirely authorial, with no need for specific reference literature, for the electrical system and the source code, and can be used in different types of equipment with compatible sensors, making only minor adjustments.

## 8.0 Acknowledgements

Sotreq would like to thank Technician Paulo Antônio for his excellent performance in carrying out this project, his persistence and knowledge made this improvement possible. In addition to Paulo, Sotreq would like to thank the commitment of the Engineering and Product Support Team, always dedicated to that which is its vocation: Provide the best mining solutions to customers.

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